

Course

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1.1.1. Calculate Momentum

02:30

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:

- m is the mass of the object (in kilograms).
- v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

calculate...

```
1 #Type Content here...
2 mass=float(input())
3 velocity=float(input())
4 momentum=mass*velocity
5 print(f"{momentum:.2f}kgm/s")
```

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits16:21

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints:
 $1 \leq n \leq 999$

Input Format:

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

Setting up environment...

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1.1.3. Days between Two Dates

07:42

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format *YYYY – MM – DD*.
- The second line contains the second date in the format *YYYY – MM – DD*.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDay...

```
1 from datetime import datetime
2 date1 = input().strip()
3 date2 = input().strip()
4 d1 = datetime.strptime(date1, "%Y-%m-%d")
5 d2 = datetime.strptime(date2, "%Y-%m-%d")
6 difference = (d2-d1).days
7 print(difference)
8
```

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1.1.4. Reverse a Number09:59

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

```
1
2 num = int(input())
3 reversed_num = int(str(num)[::-1])
4 print(reversed_num)
5
6
7
```

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1.1.5. Multiplication Table

02:48

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

```
1
2
3
4
5
6
7
8
9
10

num=int(input())
for i in range(1,11):
    print(f"{num} x {i} = {num*i}")
```

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1.2.1. Pass or Fail04:37

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:
If the student passes all courses:

passorFa...SUBMIT

```
1 marks = list(map(int, input().split()))
2
3 if len(marks) != n:
4     print("Invalid input")
5 elif any(m < 40 for m in marks):
6     print("Fail")
7 else:
8     percentage = sum(marks) / n
9     print(f"Aggregate Percentage: {percentage:.2f}")
10
11 if percentage > 75:
12     grade = "Distinction"
13 elif percentage >= 60:
14     grade = "First Division"
15 elif percentage >= 50:
16     grade = "Second Division"
17 else:
18     grade = "Third Division"
```

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1.2.2. Fibonacci Series

03:09

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci....

```
1  def fibonacci(n):
2
3
4  if n==1:
5      return 0
6  elif n==2:
7      return 1
8  else:
9      return fibonacci(n-1)+fibonacci(n-2)
10
11     return fibonacci(n-1)+fibonacci(n-2)
12
13
14 n = int(input())
15 for i in range(1, n+1):
16     print(fibonacci(i), end=" ")
17
```

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1.2.3. Pattern - 108:04

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

```
1 #Input: number of rows
2 n = int(input())
3
4 #Loop to print pattern
5 for i in range(1, n + 1):
6     print("*" * i)
```

Terminal Test cases

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1.2.4. Pattern - 200:47

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

The input is an integer, representing the number of rows in the pattern.

Output Format:

The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n=int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
6
7
```

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1.2.4. Pattern - 200:47

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Input Format:

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Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

```
1 n=int(input())
2 for i in range(1,n+1):
3     for j in range(1,i+1):
4         print(j,end=" ")
5     print()
6
7
```

TerminalTest cases

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